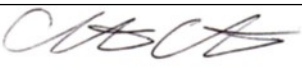


Guideline Title: Analgesia and Sedation Management

Guideline Number: GUID-3203-G005

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 2/7/2017, 6/8/2021	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To provide adequate pain relief to patients in a safe and compassionate manner.
- II. **DEFINITIONS:** None
- III. **DEPARTMENT GUIDELINE:**
 - A. Assess patient for pain using an appropriate pain scale based on patient's clinical condition and age: 1 (minimal) to 10 (severe) objective number scale for adults, Adult Nonverbal Pain Scale (NVPS), Wong Baker Faces for pediatric, FLAAC (Face, Legs, Activity, Cry, Consolability scale) for infants.
 - B. Attempt suitable non-pharmacological interventions when possible, prior to the administration of medications.
 - C. Use caution with all pharmacological agents as they may cause respiratory depression and hypotension.
 - D. Consider ETCO2 monitoring and record readings.
 - i. End Tidal CO2 monitoring must be used on all intubated patients.
 - ii. End Tidal CO2 monitoring should be used all non-intubated patients with a GCS < 14 who receive sedation or analgesia.
 - E. Monitor blood glucose levels.
 - F. Ensure reversal agents are available to the administration of opiate medications
 - i. **Naloxone:**
 1. **Adults: 0.4-2 mg IV/IO/IM/SQ or 2 mg IN** (divided equally between nostrils). Repeat as needed every 3 minutes
 2. **Pediatrics:**
 - a. < 5 yo OR < 20 kg: **0.1 mg/kg IV/IO/IM/SQ/IN q2-3 min PRN**. Maximum single dose 2 mg. Repeat as needed every 3 minutes
 - b. > 5 yrs. OR > 20 kg: **2 mg IV/IO/IM/SQ/IN** (divide IN doses equally between nostrils). Repeat as needed every 3 minutes

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- G. Consider intranasal administration in patients with seizures, burns, pediatric patients, or those with a history of autism.
- H. Consider the following medications for analgesia management:
 - i. **Fentanyl 0.5-2 mcg/kg slow IV/IO**, may repeat every 10 minutes
 - ii. **Fentanyl Intra-nasal (IN)**
 - 1. Adults: **100 mcg IN**. Administer half dose in each nostril.
 - 2. Peds: **1.5 mcg/kg IN**. Administer half dose in each nostril. Maximum dose 100 mcg.
 - iii. **Morphine**
 - 1. Adults: **0.1 mg/kg slow IV/IO**, may repeat every 10 minutes PRN
 - 2. Peds: **0.05-0.2 mg/kg slow IV/IO**, may repeat every 10 minutes PRN
 - iv. **Ketamine**
 - 1. Adults: **0.1-0.3 mg/kg slow IV/IO**, may repeat every 10 minutes PRN, maximum total dose of 50 mg
 - 2. Peds: **0.1-0.3 mg/kg slow IV/IO**, may repeat every 10 minutes PRN, maximum total dose of 25 mg
- I. Consider the following medications for sedation management:
 - i. **Midazolam 0.05-0.1 mg/kg IV/IO**, may repeat every 3-5 minutes PRN, maximum dose 5 mg
 - ii. **Midazolam Intranasal**
 - 1. Adults: **5 mg IN**. Administer half dose in each nostril.
 - 2. Peds: **0.2 mg/kg**. Administer half dose in each nostril. Maximum dose 5 mg.
 - iii. **Ketamine**
 - 1. Adults: **1-2 mg/kg slow IV/IO**. May repeat every 10 minutes PRN. Maximum total dose 200 mg.
 - 2. Peds: **1-2 mg/kg slow IV/IO**, may repeat every 10 minutes PRN. Maximum total dose of 200 mg.
 - 3. Monitor for tachydysrhythmias
 - 4. Consider **Midazolam 0.05-0.1 mg/kg IV** for signs of emergence reaction (agitation).

IV. DOCUMENTATION:

- A. EMS Charts

V. REFERENCES: